Ansh Sudani

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Education

Queen's University Sep. 2023 – Dec. 2027
Bachelor of Computing (Honours), Specialization in Software Design *Kingston, ON*

- **Relevant Coursework:** Data Structures, Algorithms I, Operating Systems, Software Architecture, Software Quality Assurance, Software Specifications, Formal Methods in Software Engineering, Coding AI for Business

Experience

Industry Development Coordinator Oct. 2026 – Present
Queen's Computing Students' Association (COMPSA) *Kingston, ON*

- Plan and execute career-focused events, workshops, and networking opportunities connecting Queen's Computing students with industry professionals.
- Conduct outreach to companies, alumni, and speakers to build external partnerships and bring industry perspectives to COMPSA programming.
- Collaborate with industry development director and coordinators, taking ownership of portfolio projects throughout the year.

Teaching Assistant Sep. 2025 – Dec. 2025
Smith Engineering, Queen's University | C *Kingston, ON*

- Supported the delivery of APSC 141 (Introduction to Computer Programming) for 750+ first-year engineering students.
- Led lab sessions teaching foundational C concepts, including loops, conditionals, functions, and data structures.
- Graded assignments, quizzes, and exams to ensure alignment with course learning outcomes.

Projects

LeetCoach | *Python, Flask, SQLite, Anthropic API*

- Built and deployed a full-stack interview-prep tool that uses a graduated Socratic hint system to guide users toward LeetCode solutions without revealing them outright.
- Designed a four-tier hint escalation system (guiding question → named pattern → applied reasoning → pseudocode) via prompt engineering with the Anthropic API.
- Implemented persistent progress tracking with SQLite and a REST API layer to manage problem state and hint usage.

J.A.R.V.I.S. – Voice-Activated AI Assistant | *Python, WebSockets, Whisper, ElevenLabs, SQLite*

- Architected a real-time streaming pipeline using Whisper (speech-to-text) and ElevenLabs (text-to-speech) over WebSockets with sub-second latency.
- Implemented function calling for web search, local automation, and system diagnostics, with a vector database for cross-session memory.
- Designed asynchronous task execution with Python's `asyncio` to handle concurrent voice input and background commands.

Technical Skills

Languages: Python, Java, C, SQL, Bash, HTML/CSS
Frameworks & Libraries: Flask, asyncio, WebSockets, Pandas, NumPy, Scikit-learn, Matplotlib
AI & APIs: Anthropic API, Claude Code, Whisper, ElevenLabs, REST APIs, Prompt Engineering
Tools & Platforms: Git, Linux, AWS (EC2, S3, RDS, Lambda), SQLite, Vector Databases, Railway

Certifications

AWS Certified Cloud Practitioner, Amazon Web Services (Projected) Dec. 2026
Complete Data Science Bootcamp, 365 Careers Dec. 2024
CS50: Introduction to Programming with Python, Harvard University Nov. 2024